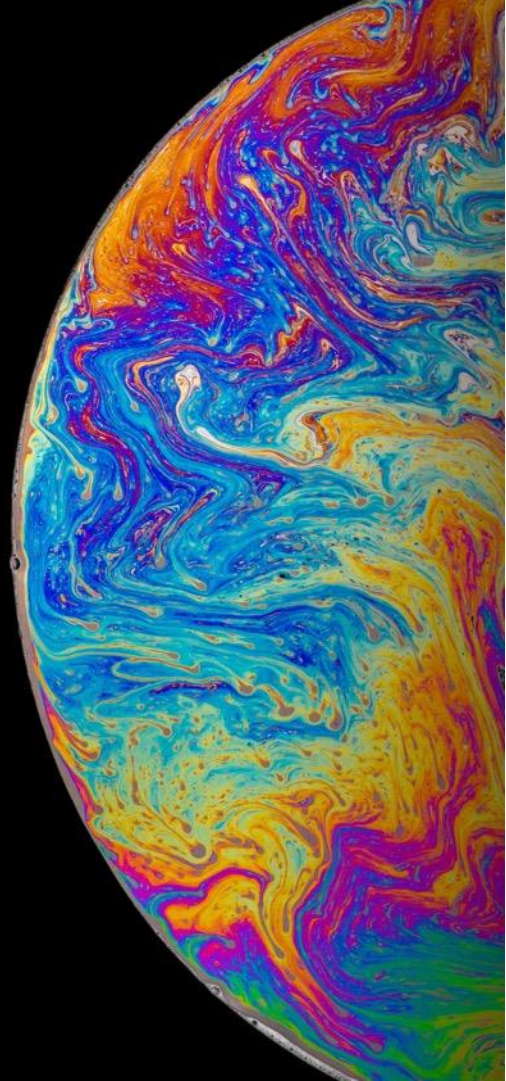
CMIP
Coupled Model Intercomparison Project

CDO and NCO for climate data

Yiwen Li

Arctic annual anomalies, 1979–2024



Python, CDO and NCO: independent routes

CDO

Climate Data Operators

Time means, anomalies and regridding
NetCDF and GRIB support

NCO

netCDF Operators

Variables, dimensions and attributes
Arithmetic and dimension averages

Each route reads the same original NetCDF files.
Python provides an independent reference and common plots.

Anatomy of a command

```
cdo selyear,1979/2014 data/historical.nc cdo/historical.nc
```

cdo	Executable
selyear,1979/2014	Operator and its parameter
data/historical.nc	Input NetCDF file
cdo/historical.nc	Output NetCDF file

The same command runs in iTerm and in the Binder notebook.

Use a new output filename to keep the input unchanged.

CDO: climate operations

One executable, many operators

Inspect	sinfo, showname
Select years	selyear
Join periods	mergetime
Convert units	setunit
Average space	fldmean
Average time	yearmean, timmean
Subtract	sub

Operator help: `cdo -h yearmean`

Examples:

```
cdo sinfo data/historical.nc
```

Shows time and grid information

```
cdo yearmean tas.nc annual.nc
```

One time mean per calendar year

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NCO: variables and dimensions

Named tools for different operations

```
ncks      Select and inspect
ncrcat    Join time records
ncap2     Calculate variables
ncwa      Average dimensions
ncra      Average time records
ncdiff    Subtract files
ncatted   Edit attributes
```

NetCDF stores variables, dimensions and descriptive attributes in one file.

Examples:

```
ncks -v tas -d lat,65.0,90.0 \
  tas.nc arctic.nc
```

Keep tas north of 65°N

```
ncwa -a lat,lon \
  arctic.nc mean.nc
```

Arithmetic spatial mean

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Chaining CDO operators

```
cdo timmean -selyear,1981/2010 \
  cdo/annual_temperature.nc \
  cdo/reference.nc
```

1 Select 1981–2010

2 Average the 30 annual values

3 Write one reference value

One command replaces two processing steps.

Processing order

The rightmost operator acts first.

Select the baseline years, then average them.

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The annual anomaly definition

ACCESS-CM2 historical + SSP5-8.5 r1i1p1f1

1979–2024 analysis

552 monthly fields become 46 annual values
Arctic: 65–90°N, all longitudes
Equal grid-point weights and equal monthly weights

1981–2010 reference

Annual anomaly = annual temperature – reference mean

Complete calendar years and fixed weights make this annual shortcut valid.

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Three files, one numerical comparison

Annual anomalies in °C, relative to 1981–2010

Python	python/arctic_annual.nc	tas_anom
CDO	cdo/annual_anomaly.nc	tas
NCO	nco/annual_anomaly.nc	tas

The comparison script reads these files explicitly.

46 matching years in each route

CDO vs Python
NCO vs Python

The same plot scale lets us compare the curves and climate stripes.

Local demonstration in iTerm

One command, one result to inspect.

First look at the data

```
cdo sinfo data/historical.nc
cdo sinfo data/ssp585.nc
```

The same lesson on Binder

https://mybinder.org/v2/gh/YiwenLiEven/Fresh-Eyes-CDO-NCO-Workshop/db8212ef0e4e5483b0c47d392a8191a149e7fdc7?urlpath=lab/tree/notebooks/05_annual_stripes_short_demo.ipynb

Open this lesson on Binder

Quick reference and documentation

CDO

sinfo	Inspect data
selyear	Select years
mergetime	Join time periods
fldmean	Spatial mean
yearmean	Annual means
timmean	Time mean

NCO

ncks	Select / inspect
ncrcat	Join records
ncap2	Arithmetic
ncwa	Dimension means
ncra	Record means
ncdiff	Subtract files
ncatted	Edit attributes

CDO documentation

code.mpimet.mpg.de/projects/cdo

NCO User Guide

nco.sourceforge.net/nco.html

The next hands-on session combines CDO and NCO in one shorter workflow.

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